Associate of Engineering

CCIT4080 Project on Knowledge Products Development

Project Guidelines – 2026/27 (Semester 1)

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(1) Introduction

Project-Based Learning in Engineering Education

Project-based learning holds an important place in engineering education because it provides students with opportunities to integrate theoretical knowledge with practical application while addressing realistic and often open-ended problems. It enables students to develop not only technical knowledge and problem-solving skills, but also important professional capabilities such as teamwork, communication, project management, decision-making, and independent learning.

Unlike lecture-based courses, where much of the required knowledge and techniques may be introduced first through lectures, tutorials, or laboratory exercises before being assessed, project-based learning requires students to take a more active role in determining what they need to learn and how to apply that knowledge.

Students should therefore expect to encounter problems, technologies, methods, tools, or technical knowledge that have not been directly taught in previous courses. Identifying knowledge gaps, locating relevant information, learning unfamiliar concepts or skills, evaluating potential approaches, and applying newly acquired knowledge to the project are important parts of the learning process.

The project supervisor provides guidance, feedback, and academic supervision. However, students are expected to progressively develop greater independence in investigating problems, making appropriate engineering decisions, overcoming technical difficulties, and evaluating the effectiveness of their chosen approaches.

Project-based learning also emphasises the development process rather than solely the final product. Students are expected to demonstrate a logical progression from identifying and understanding a problem through investigation, design, and development to testing, evaluation, refinement, and professional communication of the resulting work.

CCIT4080 as the Final Year Project

CCIT4080 Project on Knowledge Products Development is regarded as the **Final Year Project** of the Associate of Engineering (AENG) programme.

The assessments in this course are therefore intended to demonstrate the comprehensive knowledge and skills accumulated in previous courses, as well as new knowledge and skills acquired independently during the project.

Students should expect the academic and technical standards of this course to be comparatively high.

The project should demonstrate not merely completion of assigned tasks, but also the ability to integrate knowledge, investigate unfamiliar problems, make appropriate engineering decisions, develop and evaluate solutions, and communicate the resulting work professionally.

The course is intended to prepare students for further education and professional careers, where autonomy, the ability to address complex challenges, independent learning, engineering judgement, and the ability to meet high standards are essential.

Students are expected to approach their project diligently, apply their accumulated knowledge appropriately, acquire additional knowledge and skills where necessary, and demonstrate the ability to address real-world engineering problems.

As CCIT4080 forms part of the graduation requirements of the AENG programme, failing this course generally means the student has not fulfilled the graduation requirements.

Course Aims and Objectives

The course aims to provide students with an opportunity to integrate and apply knowledge and skills acquired throughout the AENG programme while developing greater independence in engineering problem-solving and project work.

In particular, the course aims to:

- **Enhance Understanding**: Deepen students' comprehension of the intricate relationships between engineering solutions and real-world problems.
- **Integrate Theory and Practice**: Enable students to apply theoretical frameworks and empirical methods in engineering problem-solving.
- **Develop Problem-Solving and Engineering Judgement**: Develop students' ability to define problems, investigate possible solutions, compare alternatives, make appropriate technical decisions, and justify the approaches adopted.
- **Promote Initiative and Autonomy**: Provide an environment in which students take increasing responsibility for project planning, technical investigation, learning, decision-making, and execution under the guidance of a project supervisor.
- **Develop Project and Teamwork Skills**: Strengthen students' abilities in project planning, time and resource management, teamwork, coordination, responsibility allocation, and professional communication.
- **Develop Evaluation and Reflective Skills**: Enable students to test and evaluate project outcomes, identify limitations, consider possible improvements, and critically reflect on both the project process and their own development.

By the end of the course, students should be able to define an appropriate project scope, establish meaningful objectives and requirements, develop and implement suitable approaches, produce and evaluate a technically sound prototype or solution, and communicate the project process and outcomes professionally.

Independent Project Work

Students should not limit their project work to the designated project sessions. The scheduled sessions provide opportunities for group collaboration, project development, supervisor consultation, progress review, and use of relevant facilities; they do not represent the total amount of time expected for completion of the project.

Students are expected to undertake substantial project work independently outside the scheduled sessions. Depending on the nature of the project, this may include background research, technical study, system design, programming, hardware construction, experimentation, testing, data collection, troubleshooting, documentation, group meetings, and preparation of assessment materials.

The supervisor's role is primarily advisory and supervisory rather than instructional. Students should not expect the supervisor to provide step-by-step solutions to technical problems or to teach all knowledge required for the project. Where unfamiliar problems or technologies are encountered, students are expected to investigate appropriate sources, acquire the necessary knowledge or skills, explore possible solutions, and seek supervisory advice where appropriate.

Students should therefore manage their time throughout the two-semester project period and maintain continual progress. A substantial project cannot reasonably be completed only during the scheduled project sessions or through concentrated work immediately before assessment deadlines.

(2) Project Description

CCIT4080 Project on Knowledge Products Development is a two-semester project course forming an integral part of the final year of the Associate of Engineering (AENG) programme. It serves as a culmination of students' academic and practical learning by requiring them to integrate knowledge and skills acquired throughout the programme and apply them to a substantial engineering project.

Nature and Scope of the Project

The project should address a clearly identified **real-world problem, engineering need, or technically meaningful challenge**. Depending on the project scope and objectives, the proposed solution may involve hardware, software, or an integrated hardware-and-software system.

Research-oriented projects that investigate or develop knowledge within an engineering-related area may also be considered. However, the project must involve substantive technical work appropriate to the course. **A literature review alone is not sufficient**. Students are normally expected to develop, implement, demonstrate, test, and evaluate a prototype, system, technical solution, experimental setup, or other project output appropriate to the approved project scope.

The scope of the project should be sufficiently substantial to demonstrate the knowledge, skills, problem-solving ability, and independence expected of a Final Year Project, while remaining realistically achievable within the two-semester project period and the resources reasonably available to the group.

The project scope and objectives may be refined during the course as students gain a better understanding of the problem, technical constraints, available resources, and results obtained. Any significant changes should be discussed with the project supervisor and appropriately documented.

Project Initiation and Topic Development

Following group formation, students are expected to collaboratively propose a project topic and conduct preliminary investigation before finalising the project direction.

The initial project proposal should establish, as far as reasonably possible at that stage:

- the problem or need to be addressed;
- the background and significance of the problem;
- the intended project goals and objectives;
- the proposed solution or general technical direction;
- preliminary feasibility considerations;
- expected project outputs;

- major resources required;
- responsibilities among group members; and
- an initial project schedule.

Project ideas should be supported by an appropriate preliminary investigation rather than based solely on assumptions. Students should consider relevant existing systems, products, technologies, theories, methods, technical literature, and other sources appropriate to the nature of the project.

The proposed topic and scope will be discussed with the project supervisor. Students may be required to refine the problem definition, objectives, technical approach, project scale, or expected output to ensure that the proposed work is feasible and appropriate for the course.

Project Development Process

Although individual projects may follow different development processes, CCIT4080 projects should normally demonstrate logical progression **from problem investigation to development and evaluation**. Students should be able to show how major engineering decisions and project activities relate to the original problem and project objectives.

The following components are generally expected.

- **Problem Investigation and Background Study**

  Students should first develop a sufficiently detailed understanding of the problem being addressed, including its context, causes, users or stakeholders where applicable, existing constraints, and the reasons why a solution or an improvement is needed.

  Relevant statements and assumptions should, where appropriate, be supported by credible evidence such as academic literature, technical documentation, industry information, market information, standards, experimental observations, or other appropriate sources.

- **Review of Existing Knowledge and Solutions**

  Students should investigate relevant theories, technologies, methods, designs, systems, products, services, or other existing solutions related to the project.

  The purpose of the review is not merely to describe what already exists. Students should analyse and compare relevant approaches, identify their strengths and limitations, and use the findings to inform the requirements and technical decisions of their own project.

  Depending on the nature of the project, this review may include academic literature, existing commercial products or services, engineering designs, algorithms, hardware components, software frameworks, technical standards, or other relevant material.

- **Requirements and Success Criteria**

  Based on the problem investigation and review, students should establish appropriate requirements for the proposed solution. These may include:

  - Functional requirements, describing what the system or solution is expected to do; and
  - Non-functional requirements, describing relevant performance, reliability, usability, safety, cost, resource, compatibility, or other constraints.

  Requirements should be sufficiently clear and, where reasonably possible, **measurable and testable**. Students should also establish appropriate success criteria, test conditions, performance metrics, or other means to determine whether the project objectives and requirements have been achieved.

- **Design**

  The **Design** describes **what solution is proposed and how it is intended to work**.

  Students should develop an appropriate overall system or solution architecture and explain how its major components, subsystems, modules, or processes interact. Appropriate engineering diagrams and technical representations should be used where applicable, such as system architecture diagrams, circuit diagrams, flowcharts, state diagrams, mechanical drawings, interface diagrams, data-flow diagrams, or software structures.

  Major design decisions should be technically justified. Where reasonable alternatives exist, students should demonstrate that appropriate options have been considered and explain why the selected approach is suitable for the project's objectives, requirements, constraints, and available resources.

- **Methodology**

  The **Methodology** describes **how the project will be developed, investigated, verified, and evaluated**.

  Students should explain the approaches and methods adopted to carry out the project and, where appropriate, compare alternative approaches before justifying the selected method.

  Depending on the project, the methodology may include development strategies, experimental procedures, data-collection methods, calibration methods, algorithm-development approaches, prototyping procedures, verification methods, testing strategies, and evaluation criteria.

The methodology should therefore provide a clear link between the project objectives and the methods used to determine whether those objectives have been achieved.

- **Implementation and Prototype Development**

  The **Implementation** describes **what has been developed, constructed, programmed, configured, integrated, or tested**.

  Students are expected to translate the proposed design into a working prototype, system, solution, or other approved project output. The implementation should demonstrate meaningful technical work and appropriate application of engineering knowledge and skills.

  Students should document important stages of development, including significant technical decisions, modifications, difficulties encountered, troubleshooting performed, and solutions adopted.

  Where appropriate, evidence of implementation may include photographs, screenshots, source-code records, circuit measurements, experimental records, fabrication outputs, version histories, demonstration records, or other relevant project artefacts.

- **Testing, Evaluation, and Refinement**

  Testing and evaluation are essential parts of the project rather than activities to be performed only after development has finished.

  Students should develop suitable test cases, experiments, measurement procedures, or other evaluation methods based on the project's objectives, requirements, and success criteria. Testing should, where applicable:

  - identify what is being tested;
  - describe the test conditions and procedures;
  - define the expected or acceptance criteria;
  - record the results obtained; and
  - compare the actual results with the intended objectives or requirements.

  Students should analyse unexpected results, limitations, failures, and discrepancies rather than simply reporting successful outcomes.

  Findings from testing and evaluation should be used to refine the design or implementation where appropriate. The final project should therefore demonstrate not only that a solution was produced, but also that its performance and limitations were critically evaluated.

- **Project Management and Continuous Development**

  Effective project management is an important part of CCIT4080. Students are expected to organise tasks, allocate responsibilities, manage resources, monitor risks, and maintain appropriate progress throughout the two-semester project period.

  The project schedule should be treated as a working project-management tool rather than merely an assessment requirement. Students should regularly compare planned progress with actual progress, identify delays or changes, explain significant differences, and revise the schedule or allocation of resources where appropriate.

  Similarly, responsibilities among group members should be sufficiently clear to allow both the group and the supervisor to identify individual contributions and monitor whether project work is progressing effectively.

  Assessment of the project therefore considers the development process and continual progress, rather than only the quality of the final deliverable. Extended periods of inactivity, insufficient evidence of development, or substantial work being postponed until immediately before assessment deadlines may negatively affect the relevant assessments even if a final prototype is eventually produced.

  Students should maintain appropriate records throughout the project to support project management, communication, assessment, and subsequent reporting.

(3) Project Group Formation

Group Size and Project Expectations

The **standard project group size is 3 to 5 members**. This group size serves as the normal benchmark for the project scope, workload, assessment requirements, and expected level of contribution.

Where a group has only **one to two members** after the group-formation process, it will normally be expected to meet the **same academic and technical standards and assessment requirements** as groups of the standard size. A smaller group does not automatically receive a reduction in project scope, workload, assessment requirements, or marking expectations.

Groups of **more than five members** are generally not encouraged and require approval from the instructor and Course Coordinator. Where such a group is approved, the project scope, technical depth, amount of development and testing, and division of responsibilities should reasonably reflect the **additional manpower available to the group**.

Each member, regardless of group size, is expected to make a **fair, meaningful, and identifiable contribution** to the project.

Group-Size Mark Adjustment

To account for the potential additional manpower available to groups exceeding the standard group size, a **group-size mark adjustment** will normally be applied.

For each additional member beyond five, 5 percentage points will be deducted from each member's final course mark, based on a 100-mark course score.

For example:

- 5 members: no adjustment
- 6 members: −5 percentage points
- 7 members: −10 percentage points

The adjustment applies to the **overall course mark**, rather than to any individual assessment component. For example, a final course mark of 78/100 would become 73/100 after a 5-percentage-point adjustment.

Group Formation Process

Students are expected to form their own project groups during the first two lessons.

Within one week after the end of Lesson 2, students are required to submit their draft group formation to the instructor.

Students who have not yet formed or joined a group by the deadline remain responsible for making the necessary arrangements themselves. **The instructor and Course Coordinator will not assign students to groups, combine groups, or determine group membership on students' behalf**.

The finalised group list will normally be announced before **Lesson 3**. Once the group has been formed, students are expected to sit and work together during the designated meeting sessions.

Once confirmed, students should not independently add, remove, or exchange group members. Any subsequent change in group membership must be discussed with and approved by the instructor and/or Course Coordinator.

Role Allocation and Division of Labour

- **Group Representative**

  Each group must select a **Group Representative** who will serve as the **primary contact between the group and the project supervisor**.

  The Group Representative is primarily responsible for facilitating communication between the group and the supervisor, including conveying relevant information and communicating with the supervisor on the group's behalf.

  The role of Group Representative **does not necessarily imply the role of a group leader**, nor does it give the representative greater authority over other group members. Groups may organise their internal leadership, coordination, and task management in a manner appropriate to their project, and these responsibilities may be undertaken by the Group Representative or by other member(s).

  All group members remain responsible for their own participation, assigned work, awareness of course requirements and deadlines, and contribution to the overall project.

- **Division of Labour**

  Group members should establish clear roles and responsibilities according to the nature of the project, the work required, and the abilities and strengths of individual members.

The allocation of work should be appropriately documented and updated where necessary through project records such as the project schedule, Logbook, meeting records, progress reports, or other relevant documentation.

A balanced contribution does not necessarily mean an equal division of workload. Different members may undertake varying types and amounts of work depending on their roles, technical strengths, responsibilities, and the project's needs. However, each member is expected to make a fair and meaningful contribution that can be identified and demonstrated.

Although responsibilities may be divided among members, students should maintain a reasonable understanding of the overall project and how their own work contributes to the complete system or solution. The allocation of a particular task to one member does not remove the collective responsibility of the group to monitor project progress and ensure that required deliverables are completed.

Individual contribution will be considered through the relevant individual assessment components, project-session observations, Peer Evaluation(s), Self-Reflection Report, project records, and other appropriate evidence as specified in the assessment arrangements.

Supervisor Involvement

The project supervisor will provide consultative guidance and monitor the overall progress of the project.

The supervisor may provide feedback on matters such as project scope, feasibility, technical direction, project management, implementation, testing, and evaluation. However, consistent with the project-based nature of CCIT4080, **the supervisor's role is primarily advisory rather than instructional**.

Students are expected to investigate technical problems, acquire necessary knowledge and skills, consider alternative approaches, and make appropriate engineering decisions with increasing independence.

(4) Bi-Weekly Project Sessions

The bi-weekly project sessions provide regular contact time for **project development, group collaboration, supervisor consultation, and progress review**. Students are expected to make productive use of the entire scheduled session according to the current needs and stage of their project.

As stated in Section (1), the scheduled project sessions represent only part of the time required for the project. Students are expected to undertake substantial project work independently outside these sessions and use the scheduled sessions to continue development, coordinate with group members, review progress, and obtain supervisory feedback.

Attendance

Attendance at the bi-weekly project sessions is **mandatory**.

Students are responsible for recording their attendance using the method(s) provided by the instructor and the College's attendance system. Attendance records will be used to monitor participation throughout the course.

Students are expected to attend the entire scheduled project session. **Prolonged lateness, early departure, or absence may result in a deduction from the final course mark**. In particular, **attendance below 70% throughout the course, without a valid reason and appropriate supporting documentation, will result in a significant mark deduction**.

The consultation period allocated to each group represents only part of the scheduled project session. Students who attend only their group's consultation period—for example, by arriving shortly before the consultation and leaving immediately afterwards—will normally be considered **absent from that project session**.

Participation and Collaboration

Students are expected to participate actively throughout the project sessions. Attendance alone does not necessarily demonstrate meaningful participation.

Depending on the nature and current stage of the project, appropriate activities may include:

- discussing and reviewing project progress with group members;
- coordinating tasks and responsibilities;
- conducting research or technical investigation;
- developing or refining system designs;
- programming, constructing, integrating, or troubleshooting project components;

- conducting experiments or testing;
- analysing data or test results;
- preparing or reviewing technical documentation;
- updating project schedules, project records, or the Logbook;
- preparing materials or questions for supervisor consultation; and
- discussing problems, possible solutions, and subsequent project activities.

The project sessions also provide an important opportunity for group members to meet in person, exchange updates, resolve coordination issues, and work collaboratively on problems that may be more difficult to address independently.

As the scheduled sessions are shared among multiple project groups, students should expect periods during which the supervisor is consulting other groups. Students are expected to continue project-related activities during these periods and make appropriate use of the available session time.

Starting from Lesson 3, after group formation has been finalised, groups are expected to attend each session with relevant project materials and evidence of progress appropriate to their current stage of development. These may include written ideas, research findings, diagrams, calculations, schedules, source code, hardware, prototypes, test results, data, photographs, videos, screenshots, draft reports, or other relevant materials.

Supervisor Consultation

The project sessions provide regular opportunities for groups to consult their project supervisor.

Supervisor consultation may include:

- reporting and reviewing current project progress;
- discussing project scope, requirements, or technical direction;
- reviewing designs, prototypes, implementation, or testing;
- discussing problems or difficulties encountered;
- considering alternative approaches or possible improvements;
- reviewing project schedules and upcoming activities; and
- receiving feedback and guidance on subsequent project work.

Groups should prepare appropriately for consultation so that the available time can be used effectively. Students should, wherever possible, present specific work, evidence, questions, difficulties, or decisions for discussion rather than providing only general statements about progress.

The supervisor will provide advice, feedback, and guidance where appropriate. However, consistent with the project-based nature of CCIT4080, students remain responsible for investigating problems, considering possible solutions, carrying out the necessary project work, and making appropriate decisions.

Progress Reporting and Individual Contribution

Groups may be required to provide formal or informal progress updates during the project sessions. These updates are part of the project's continuous monitoring and allow the supervisor to review whether the project is progressing appropriately.

Each student is expected not only to attend and participate in the sessions but also to be able to **explain their own responsibilities, work completed, current progress, and contribution to the overall project**.

As some forms of project work may be completed outside the scheduled sessions or may not be readily demonstrated during consultation, students are encouraged to maintain appropriate supporting evidence. Depending on the nature of the work, this may include:

- photographs or videos;
- screenshots or screen recordings;
- source-code or version-control records;
- design diagrams or calculations;
- experimental records or test data;
- prototype-development records;
- meeting records or task records; and
- other relevant project artefacts.

Such evidence may be used to support students' explanations of their progress and contribution.

Observations made during the project sessions, together with relevant project records, submitted assessments, Peer Evaluation(s), the Individual Self-Reflection Report, and other appropriate evidence, may be considered when assessing individual contribution and relevant project assessment components.

A student who repeatedly demonstrates little or no meaningful progress or contribution may receive a lower individual mark in the relevant assessment components, even where the group as a whole produces satisfactory or better project outcomes.

(5) Schedule and Assessment

Duration and Course Schedule

CCIT4080 Project on Knowledge Products Development spans **two semesters**, comprising **12 scheduled project sessions (6 sessions per semester)**, and carries **3 credits**.

The project is developed continuously across both semesters. The scheduled project sessions, assessment milestones, and submission deadlines are intended to support the progressive development of the project from initial problem investigation and proposal, through design and implementation, to testing, evaluation, and final presentation of the completed work.

Students are responsible for monitoring the course schedule, SOUL announcements, and any subsequent updates issued by the Course Coordinator or project supervisor.

Assessment Overview

The overall course assessment comprises the following components:

Assessment Component	Weighting
Project Proposal Presentation	5%
Interim Report – 1st Submission	5%
Interim Report – 2nd Submission	5%
Individual Self-Reflection Report	5%
Peer Evaluation(s) – 5 evaluations × 1%	5%
Project Final Report	20%
Project Deliverable	35%
Project Oral Presentation	15%
Logbook – Semester 1	2.5%
Logbook – Semester 2	2.5%
Total	**100%**

Detailed requirements and marking criteria for individual assessment components are provided in **Section (6) Assessment Details** and the corresponding marking rubrics.

Where applicable, **course-level mark adjustments** specified elsewhere in these Guidelines will be applied to the overall course mark. These may include, for example, the group-size adjustment described in Section (3) and attendance-related adjustments described in Section (4).

General Assessment Principles

CCIT4080 assesses both the project development process and the resulting project outcomes. Assessment is therefore not based solely on whether a final prototype or product has been completed.

Depending on the assessment component, consideration may be given to:

- understanding and investigation of the problem;
- review and appropriate use of relevant existing knowledge, technologies, systems, products, or solutions;
- definition of meaningful and testable project objectives and requirements;
- appropriateness and justification of engineering and technical decisions;
- quality and depth of system or solution design;
- implementation and technical development;
- project progress throughout the two-semester period;
- testing, evaluation, analysis, and refinement of the project;
- quality and functionality of the final prototype, solution, or other approved project output;
- project planning, scheduling, and resource management;
- quality and professionalism of technical documentation and presentation;
- teamwork, participation, and communication; and
- identifiable individual contribution.

Students are expected to support significant claims concerning project progress, implementation, performance, testing, or achievement with **appropriate evidence**. Depending on the nature of the project, such evidence may include technical diagrams, calculations, source-code records, photographs, videos, screenshots, prototype records, test procedures, experimental data, test results, version histories, project records, or other appropriate artefacts.

The quality of the evidence and analysis is important. Merely stating that a feature has been completed, a system is effective, a project is "on schedule", or an objective has been achieved does not by itself demonstrate the claim.

Where an assessment requires review of project progress, students should distinguish clearly among:

- completed work;
- work currently in progress; and
- planned or remaining work.

Similarly, where scheduling or resource allocation is assessed, students should review **planned versus actual progress or usage**, explain significant differences, and revise subsequent plans where appropriate.

Academic Calendar and Project Session Schedule

Week #	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1			16/09	17/09	18/09	19/09
2	21/09	22/09	23/09	24/09	No Class	No Class
3	28/09	29/09	30/09	No Class	02/10	03/10
4	05/10	06/10	07/10	No Class	09/10	10/10
5	12/10	13/10	14/10	15/10	16/10	17/10
6	No Class	20/10	21/10	22/10	23/10	24/10
7	26/10	27/10	28/10	29/10	30/10	31/10
8	02/11	03/11	04/11	05/11	06/11	07/11
9	09/11	10/11	11/11	12/11	13/11	14/11
10	16/11	17/11	18/11	19/11	20/11	21/11
11	23/11	24/11	25/11	26/11	27/11	28/11
12	30/11	01/12	02/12	03/12	04/12	05/12
13	07/12	08/12	09/12	10/12	11/12	12/12
14	14/12					
07 Oct: Teaching Thursday Schedule						

Academic Calendar – 2026/27 Semester 1

Wed.			Thu.		Fri.						Sat.	
08:30 – 11:30	16:00 – 19:00		10:00 – 13:00		10:00 – 13:00		13:00 – 16:00		16:00 – 19:00		08:30 – 11:30	
CL13	CL11	CL12	CL01	CL02	CL07	CL08	CL05	CL06	CL09	CL10	CL03	CL04

Bi-Weekly Project Session Schedule – 2026/27 Semester 1

Time \ Day	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
08:30 -	CCIT4081-CL01	CCIT4081-CL04	CCIT4080-CL13			CCIT4080-CL03
10:00 -				CCIT4080-CL01	CCIT4080-CL07	CCIT4080-CL04
11:30 -				CCIT4080-CL02	CCIT4080-CL08	
13:00 -	CCIT4081-CL02	CCIT4081-CL05			CCIT4080-CL05	
14:30 -				CCIT4030-CL03	CCIT4080-CL06	
16:00 -			CCIT4080-CL11		CCIT4080-CL09	
17:30 -	CCIT4081-CL03		CCIT4080-CL12		CCIT4080-CL10	

Note: The greyed-out time slots indicate the likelihood that KEC701 will be idle/occupied. The actual availability will be confirmed by the AENG Teaching Team. Once confirmed, some empty time slots may be made available to Year 2 students for project-related purposes.

KEC701 Engineering Teaching Laboratory Overall Schedule – 2026/27 Semester 1

Week #	Monday	Tuesday	Wednesday	Thursday	Friday
1					22/01
2	25/01	26/01	27/01	28/01	29/01
3	01/02	02/02	03/02	04/02	No Class
4	No Class	No Class	No Class	11/02	12/02
5	15/02	16/02	17/02	18/02	19/02
6	22/02	23/02	24/02	25/02	26/02
7	01/03	02/03	03/03	04/03	05/03
8	08/03	09/03	10/03	11/03	12/03
9	15/03	No Class	17/03	18/03	19/03
10	22/03	23/03	24/03	25/03	No Class
11	No Class	30/03	31/03	01/04	02/04
12	No Class	06/04	07/04	08/04	09/04
13	12/04	13/04	14/04	15/04	16/04
14	19/04	20/04	21/04	22/04	23/04
15	26/04	27/04			
08 April: Teaching Monday Schedule					

Academic Calendar – 2026/27 Semester 2 (TBC*)

Monday		Tuesday	Wednesday		Thursday		Friday	

Bi-Weekly Project Session Schedule – 2026/27 Semester 2 (TBC*)

Day / Time	Monday	Tuesday	Wednesday	Thursday	Friday
08:30 -					
10:00 -					
11:30 -					
13:00 -					
14:30 -					
16:00 -					
17:30 -					

KEC701 Engineering Teaching Laboratory Overall Schedule – 2026/27 Semester 2 (TBC*)

Semester 1		
Lesson	**Activity/Milestone**	**Weighting**
1 & 2	**Course Introduction** - Overview of course objectives, expectations, and assessment criteria. **Group Formation** - Students are expected to form groups comprising 3 to 5 members. **Topic Selection** - Groups should begin brainstorming and selecting a project topic for approval. **Lesson 2 Milestone - Group Formation** - By the end of Lesson 2, students are required to submit their draft group formation to the instructor. **Pre-Lesson 3 Milestone - Group Confirmation** - One week following Lesson 2, the instructor will post the finalized group list on SOUL. Starting from Lesson 3, students are required to sit and work together as a group.	
3	**Preliminary Research** - Students must engage in a preliminary study to understand the scope of their chosen problem and explore potential solutions.	
4	**Project Proposal Presentation** - Students are required to present their **Project Proposals** during this lesson. The presentation should clearly outline the problem being addressed, the proposed solution, and the plan for project execution. - Presentation slides must be submitted before the start of Lesson 4. The presentation will be recorded for assessment and review purposes. **Lesson 4 Assessment & Milestone** - The **Project Proposal Presentation**, along with the **revised slides**, will contribute **5% of the overall course grade**.	**5%**
5	**Feedback on Proposal** - The instructor will provide feedback on the project proposals, highlighting strengths and areas for improvement. **Problem Investigation** - Students should conduct a detailed examination of the problem and refine their proposed solutions based on the feedback received. The revised proposal (slides) must be submitted within one week after Lesson 5. **Interim Report Briefing** - A briefing will be conducted to outline the requirements and expectations for the upcoming **Interim Report**, ensuring students understand the key elements to include and the standards to meet.	
6	**Project Progress Presentation** - Students must present the current status and progress of their projects. The presentation should adhere to formal academic standards and cover key developments, challenges encountered, and plans for the next stages of the project. **Pre-Examination Period Assessment & Milestone** **Logbook Submission** - Groups must submit their **1st Semester Logbook** on or before 11:59 pm, Wednesday, 16th December 2026. This submission will account for **2.5% of the overall course grade**.	**2.5%**

Semester 2		
Lesson	**Activity/Milestone**	**Weighting**
1	**Pre-Semester Start Assessment & Milestone** – Students are required to submit their **Interim Report (1st Submission)** on or before 11:59 pm, Saturday, 9th January 2027. This submission will account for **5% of the overall course grade**. **Feedback on Interim Report (1st Submission)** – The instructor will provide feedback on the **Interim Report (1st Submission)** during Lesson 1, highlighting areas for improvement and offering guidance on how to enhance the report's quality. **Lesson 1 Assessment & Milestone** **Interim Report (2nd Submission):** – Groups must submit their **Interim Report (2nd Submission)** within one week after Lesson 1. This submission will account for **5% of the overall course grade**.	**5%** **5%**
2 - 5	**Assessments Briefing** – A briefing will be conducted to outline the requirements and expectations for the upcoming assessments. **Feedback on Interim Report (2nd Submission)** – The instructor will provide feedback on the **Interim Report (2nd Submission)**, highlighting areas for improvement and offering guidance on preparing the Final Report. **Project Progress & Prototype Review** – Students are required to present the current status and advancements of their projects, including updates on research, design, and development stages. – Students must also present their work-in-progress prototype (or testing evidence) for a preliminary review. This is a critical checkpoint to ensure functionality and design are on track before the final evaluation. **Failure to present a preliminary prototype or testing evidence during these lessons will result in a significant downgrade in the relevant assessments.** **Post-Lesson 4 Milestone Assessment & Milestone** **Individual Self-Reflection Report Submission:** – Students must submit their **Individual Self-Reflection Report** within one week after Lesson 4. This submission will account for **5% of the overall course grade**.	**5%**
5* - 6	**In-Class Project Final Presentation** – Students are required to present their final project outcomes during this lesson. – Due to limited contact hours, in-class presentations will focus on the project outcomes, such as the status of the prototype, testing performed, and test results. This approach is designed to gather supervisors' feedback, allowing students to refine and enhance their self-recorded final presentations. **Pre-Semester End Assessment & Milestone** – Students must submit the following items by **11:59 pm, Tuesday, 27th April 2027 (except for classes with their last lesson on 26th or 27th April 2027)**. These items contribute to the overall course grade: – **2nd Semester Logbook (2.5%)** – **(Self-Recorded) Project Presentation (15%)*** – **Project Final Report (20%)*** – **Project Deliverable (35%)***	**2.5%** **15%** **20%** **35%**
		Total: 95%#

Excluded Peer Evaluation(s) (5%)

Remark: In addition to the required SOUL submissions, each group must maintain a well-organised shared OneDrive folder containing the group's project assessment materials and supporting files for supervisor and second-examiner access. The shared folder should include the relevant group assessment submissions and supporting project materials, but should exclude individual Peer Evaluation submissions.

Detailed requirements for these assessments are provided in Section (6).

(6) Assessment Details

General Assessment Requirements

SOUL is the primary platform for formal assessment submission. Students are responsible for ensuring that the correct files are submitted in the required format and by the applicable deadline specified in Section (5) and/or subsequent course announcements.

In addition, each group must maintain the shared OneDrive folder described in Section (5) for project archiving, supervisor access, and second-examiner access where applicable.

The assessment descriptions in this section provide guidance on the **expected scope and content** of each assessment. They should be read together with the corresponding marking rubrics. The suggested headings or structures are not necessarily rigid templates; depending on the nature of the project, students may organise their work differently where an alternative structure communicates the required content more appropriately.

Assessment Based on Content and Evidence

Assessment is based on the **quality and relevance of the work actually presented**, rather than merely the presence of a section heading, diagram, table, prototype feature, or other item.

Where a required or assessed element is materially absent, the corresponding marking category may be assessed within the **Inadequate** band. Merely including a heading without meaningful relevant content does not constitute completion of that element.

Students should ensure that significant claims concerning project design, implementation, progress, functionality, performance, testing, or achievement are supported by appropriate evidence wherever applicable.

Evidence may include, depending on the nature of the project:

- academic or technical references;
- engineering calculations;
- system, circuit, mechanical, software, or other technical diagrams;
- photographs or videos of development work;
- screenshots or screen recordings;
- source-code or version-control records;
- experimental procedures and observations;
- test cases, measurement data, and test results;
- project schedules and progress records;
- meeting records and Logbook entries; and
- other relevant project artefacts.

Claims that cannot be reasonably supported by the submitted work or other available project evidence may receive limited credit.

Accuracy and Consistency of Project Reporting

Assessment materials should accurately represent the **actual status of the project**. Students should clearly distinguish between:

- work that has been completed;
- work that is currently in progress; and
- work that is planned but not yet completed.

Planned features, intended experiments, expected results, or proposed improvements should not be presented as completed achievements.

Information should also remain reasonably consistent across different parts of the project documentation. Where the project scope, technical approach, requirements, schedule, resources, or implementation has changed, the change should be clearly explained and the relevant assessment materials updated accordingly.

Use of Appropriate Sources and References

Students are expected to use appropriate and credible sources to support background information, technical decisions, engineering claims, component specifications, market comparisons, methodologies, and other externally derived information where relevant.

The quality, relevance, and appropriate use of sources may affect the assessment of the corresponding content.

All sources used should be cited consistently. **APA and IEEE citation styles are preferred for this course**, although another recognised academic citation style may be used consistently where appropriate.

Group and Individual Assessment

CCIT4080 includes both **group-based and individual assessments**.

Group-based marks reflect the overall quality of the work produced collaboratively, whereas individual marks consider the student's own contribution, involvement, technical work, communication, and other relevant evidence.

Students within the same group should therefore not assume that identical final marks will necessarily be awarded where meaningful differences in individual contribution or performance are identified. Relevant evidence may include:

- observations during project sessions;
- work demonstrated during supervisor consultation;
- submitted project materials;
- Logbook and project records;
- Peer Evaluation(s);
- Individual Self-Reflection Reports;
- presentation performance; and
- other appropriate evidence of individual contribution.

Project Proposal Presentation (5%)

The Project Proposal Presentation represents the first formal assessment of the proposed project. Its purpose is to demonstrate that the group has identified an appropriate problem, conducted sufficient preliminary investigation, developed a feasible preliminary solution, and established a realistic plan for carrying out the project over the two-semester period.

Each group should normally allocate approximately **15 minutes** for the presentation.

The presentation should be prepared for an engineering audience and should communicate the proposed project clearly and professionally. Supporting references, diagrams, preliminary calculations, comparisons, or other evidence should be used where appropriate.

The presentation should address the following areas.

- **Problem Statement, Project Goals and Objectives**

  Clearly define the real-world problem, engineering need, or technical challenge that the project intends to address.

  The problem should be sufficiently specific and should be supported, where appropriate, by credible evidence such as academic publications, technical reports, industry information, market information, standards, statistics, or other relevant sources.

  The group should explain:
  - what the identified problem is;
  - who or what is affected by the problem, where relevant;
  - why the problem is meaningful or worth addressing;
  - the overall goal of the proposed project; and
  - the specific objectives that the project intends to achieve.

  Project objectives should be sufficiently clear to guide subsequent development. Where reasonably possible, objectives should be **measurable or evaluable** at the end of the project.

  The group should therefore begin considering appropriate success criteria, performance indicators, test cases, measurements, or other methods that could later be used to determine whether the objectives have been achieved.

- **Project Feasibility and Proposed Solution**

  Present the proposed solution and explain **how it is expected to address the identified problem and achieve the project objectives**.

  The presentation should demonstrate that the group has conducted sufficient preliminary investigation to support the proposed direction rather than selecting technologies or approaches without justification.

  Where appropriate, the group should consider:
  - the overall system or solution concept;
  - major hardware and/or software components;

- relevant engineering principles, technologies, or methods;
- possible alternative approaches;
- major technical challenges or uncertainties;
- availability of required equipment, software, materials, or other resources;
- expected cost;
- technical complexity;
- time available for development; and
- major risks that could affect completion of the project.

Where multiple reasonable approaches are available, students are encouraged to briefly compare them and explain why the proposed approach is considered suitable.

At the proposal stage, the design does not need to be final. However, the group should demonstrate that the proposed solution is **technically meaningful, reasonably feasible, and sufficiently developed to provide a credible basis for subsequent project work**.

- **Schedule**

Provide a project schedule covering the major stages of work from the beginning of the project to the final project deadline in late April.

The schedule should identify appropriate:

- tasks or work packages;
- major milestones;
- dependencies where relevant;
- target completion periods;
- development and integration stages;
- testing and evaluation activities;
- preparation of assessment deliverables; and
- reasonable contingency or buffer time.

The schedule should be sufficiently detailed to allow subsequent comparison between **planned and actual progress**.

Students should avoid placing most implementation, integration, or testing activities immediately before the final submission deadline. Sufficient time should be reserved for troubleshooting, testing, evaluation, refinement, and preparation of the final assessment materials.

- **Resource Allocation**

Explain how the resources required for the project will be allocated.

This should include human resources, with the preliminary responsibilities or major work areas assigned to each group member.

Depending on the project, resource planning should also consider:

- hardware components;
- materials and fabrication;
- laboratory equipment;
- software, platforms, APIs, or cloud services;
- computing resources;
- datasets or other information resources;
- estimated expenditure or project budget; and
- other specialised resources required for development or testing.

The proposed allocation should be realistic and consistent with the project scope and schedule.

The division of work should demonstrate that each member is expected to make a fair, meaningful, and identifiable contribution, while recognising that responsibilities may change as the project develops.

- **Overall Presentation Quality**

The presentation should communicate the proposal in a clear, logical, concise, and professional manner. Assessment may consider:

- organisation and logical flow;
- clarity of explanations;
- quality and relevance of presentation materials;
- effective use of diagrams, figures, tables, and references;
- appropriateness of technical depth;
- presentation delivery;
- time management; and
- the group's ability to demonstrate an informed understanding of the proposed project.

Presentation materials should support the explanation rather than consist primarily of large amounts of text intended to be read directly from the slides.

- **Assessment Weighting**

The Project Proposal Presentation contributes 5% of the overall course grade. The corresponding assessment areas are:

Assessment Area	Weighting within Assessment
Problem Statement, Project Goals and Objectives	10%
Project Feasibility and Proposed Solution	30%
Schedule	20%
Resource Allocation	20%
Overall Presentation Quality	20%
Total	**100%**

Interim Report (10%)

The Interim Report is intended to demonstrate the **substantive development of the project beyond the proposal stage**. It should provide an evidence-based account of the group's current understanding of the problem, requirements, design, technical decisions, implementation progress, testing, project management, and remaining work.

The Interim Report consists of two assessed submissions*:

- **Interim Report – 1st Submission: 5%**
- **Interim Report – 2nd Submission: 5%**

Each group is expected to produce a professionally formatted report of **at least 1000 words**.

The 2nd Submission should be a revised and improved version of the report that meaningfully responds to the feedback provided on the 1st Submission. Revision should address technical, analytical, structural, evidential, and presentation issues where applicable, rather than being limited to minor wording or formatting changes.

The following structure is expected. Project-specific subsections may be added or reorganised where appropriate, provided that the required assessment content remains clearly identifiable.

- **Front Page**

  The Front Page should include the following information in a clear and professional hierarchy:
  - College Name;
  - Programme Name;
  - Course Code and Course Name;
  - Class Number and Group Number;
  - Project Title;
  - Assessment Item
    - i.e. **Interim Report – 1st Submission** or **Interim Report – 2nd Submission**;
  - group members' names and Student IDs (SIDs);
  - supervisor's name; and
  - submission date.

- **Abstract**

  Provide a concise summary of the project and its **current interim status**. The Abstract should normally summarise:
  - the problem being addressed;
  - the proposed solution;
  - the principal approach or methodology;
  - work completed to date;
  - significant interim findings or results, where available;
  - major work that remains outstanding; and
  - the next stage of development.

  The Abstract should accurately reflect the content of the report. Strong claims concerning implementation, performance, effectiveness, completion, or results should not be made unless they are supported by evidence presented in the main body of the report.

- **Acknowledgement (if applicable)**
  Acknowledge relevant assistance, facilities, resources, technical support, or other contributions received from persons or organisations outside the project group.

- **Table of Contents**
  Provide a properly structured Table of Contents showing the report headings and corresponding **starting page numbers**.

  The section titles and numbering shown in the Table of Contents should match the headings used in the report.

- **List of Figures**
  Provide a numbered list of all figures with their captions and corresponding page numbers.

  All figures should:
  - be numbered consistently;
  - contain meaningful captions;
  - be referred to and explained in the main text; and
  - identify or cite external sources where applicable.

- **List of Tables (if applicable)**
  Provide a numbered list of all tables with their titles and corresponding page numbers.

  All tables should be numbered consistently, clearly titled, and referred to appropriately in the main text.

- **Introduction**
  Introduce the project sufficiently for a reader who may not be familiar with the original Project Proposal.

  The Introduction should normally include:
  - project background and context;
  - the specific problem or need being addressed;
  - project goals and objectives;
  - scope and relevant constraints;
  - important theories, concepts, or terminology required to understand the project; and
  - a brief overview of the development process and current project stage.

  Project objectives should be stated clearly and, where reasonably possible, in a manner that can later be evaluated.

  Where the project scope, objectives, or technical direction have changed substantially since the Project Proposal, the changes and reasons should be clearly explained.

- **Literature Review**

  Review and critically analyse relevant literature, existing systems, products, services, technologies, theories, designs, methods, or other prior work that informs the project.

  The scope of the Literature Review will depend on the nature of the project. It may include:
  - academic literature;
  - existing commercial products or services;
  - comparable engineering systems or prototypes;
  - relevant theories and concepts;
  - engineering designs or reference architectures;
  - algorithms and technical methods;
  - hardware or software technologies;
  - standards and specifications;
  - datasheets or other authoritative technical documentation; and
  - other relevant sources.

  The Literature Review should not merely summarise existing work. It should demonstrate how the review informed the development of the project by:
  - comparing relevant alternatives;
  - identifying strengths and limitations;
  - identifying gaps or unresolved problems;
  - supporting technical assumptions;
  - justifying requirements; and
  - informing subsequent design and engineering decisions.

  Where appropriate, comparison tables or other structured forms should be used to compare existing systems, products, methods, or technical approaches.

- **Project Requirements**

  Relevant findings from the Literature Review should be translated into appropriate project requirements and design considerations. Where suitable, students should identify:
  - **Functional Requirements (FRs)**: what the proposed system or solution is required to do; and
  - **Non-Functional Requirements (NFRs)**: relevant performance, reliability, usability, safety, cost, compatibility, resource, environmental, or other constraints.

  Requirements should be sufficiently clear and, where reasonably possible, measurable and testable. Numerical targets or thresholds should be supported by appropriate sources, calculations, preliminary investigation, or other reasonable justification rather than being selected arbitrarily.

  The use of identifiers such as FR-01, FR-02, NFR-01, NFR-02, etc. is encouraged where this improves clarity and traceability.

  Students should aim to establish a logical relationship between:
  Problem and Objectives → Literature / Existing Solutions → Derived Requirements → Design and Testing

- **Design, Methodology and Implementation**

  This section should explain the proposed technical solution, the approach used to develop and evaluate it, and the work that has actually been implemented at the interim stage.

  Depending on the nature of the project, the section may be divided into the following subsections.

  - **Design**

    The Design should explain **what has been designed and how the proposed system or solution is intended to work**.

    Students should provide an appropriate overall system or solution architecture and explain how the major components, subsystems, modules, or processes interact.

    Depending on the project, appropriate content may include:
    - overall system architecture;
    - hardware and software subsystems;
    - circuit design;
    - mechanical or structural design;
    - software architecture;
    - algorithms and processing logic;
    - data structures or databases;
    - user-interface design;
    - communication interfaces;
    - control logic;
    - workflow or state transitions;
    - component selection;
    - integration between subsystems; and
    - etc.

    Appropriate engineering representations should be used where applicable, such as system block diagrams, circuit schematics, flowcharts, state diagrams, mechanical drawings, wiring diagrams, interface diagrams, software architecture diagrams, or other relevant technical figures.

    Major design and component choices should be justified. Where reasonable alternatives exist, students should compare appropriate options using relevant criteria such as performance, cost, reliability, complexity, compatibility, availability, power consumption, development time, or other project-specific constraints.

    The design should clearly respond to the requirements and findings established from the earlier sections.

  - **Methodology**

    The Methodology should explain **how the project is being developed, investigated, verified, and evaluated**.

    Depending on the project, this may include:
    - development or prototyping approach;
    - experimental procedures;
    - technical decision-making methods;
    - data-collection methods;
    - calibration procedures;
    - algorithm-development approaches;

- integration strategy;
- verification methods;
- testing procedures; and
- evaluation metrics or acceptance criteria.

Where alternative approaches or methods have been considered, students should briefly compare them and justify the selected approach.

The Methodology should not simply repeat the Design or provide a list of technologies used. It should **explain the approach and reasoning used to develop and evaluate the solution**.

- **Implementation**

The Implementation should provide an evidence-based account of **what has actually been developed, constructed, programmed, configured, integrated, or tested by the time of submission**.

Students should clearly distinguish among:
- completed work;
- work currently in progress; and
- planned or remaining work.

Depending on the project, appropriate evidence may include:
- prototype photographs;
- circuit or wiring records;
- fabrication outputs;
- screenshots of implemented software;
- source-code or version-control records;
- implemented algorithms;
- interface screenshots;
- experimental setups;
- system-integration evidence;
- measurement records;
- test logs; and
- other relevant project artefacts.

Where implementation differs from the original design or proposal, the changes and reasons should be explained.

Students should also discuss significant technical difficulties encountered, investigation or troubleshooting undertaken, solutions attempted, and the resulting outcomes.

Where preliminary testing has been conducted, the report should explain:
- what was tested;
- the test setup and conditions;
- the procedure used;
- the relevant requirement or objective;
- expected or acceptance criteria, where applicable;
- results obtained; and
- interpretation of the results.

Where meaningful testing has not yet been completed, this should be stated accurately and the planned testing approach should be described instead.

- **Scheduling and Resource Allocation**

Review the project schedule and resource allocation against the actual progress and resource usage to date.

The schedule review should identify:

- originally planned activities and milestones;
- actual completion status;
- differences between planned and actual progress;
- delays or changes and their causes;
- outstanding tasks;
- revised target dates; and
- appropriate recovery actions or contingency arrangements where necessary.

A revised schedule should be provided where the original schedule is no longer accurate.

The resource review should consider, where applicable:

- allocation of responsibilities among group members;
- equipment and laboratory resources;
- hardware components and materials;
- software and computing resources;
- APIs, cloud services, or other external services;
- expenditure and budget;
- resources already used or purchased; and
- resources still required.

Statements that a project is "on schedule", "within budget", or progressing according to plan should be supported by an appropriate comparison between planned and actual progress or resource usage.

- **Discussion**

Critically **discuss the current state of the project** based on the work and evidence presented earlier in the report. The Discussion may address:

- progress towards the project objectives;
- extent to which requirements have currently been addressed;
- effectiveness of the design or implementation to date;
- interpretation of preliminary testing or experimental results;
- technical strengths and limitations;
- problems encountered and their causes;
- trade-offs or compromises made;
- lessons learned from the development process;
- risks affecting the remaining project work; and
- priorities for the next stage.

The Discussion should analyse and interpret the project evidence rather than merely repeat earlier descriptions. Results or achievements that have not been documented in the Design, Methodology and Implementation section should not be introduced for the first time as unsupported claims in the Discussion.

As this is an Interim Report, the discussion should accurately recognise that the project remains under development.

- **Individual Progress**

Provide a separate subsection for each group member summarising their individual responsibilities and progress. Each member's subsection should include:

- a portrait photograph;
- full name and Student ID (SID);
- project role or major responsibilities;
- specific tasks completed;
- concrete outputs or deliverables produced;
- current work in progress; and
- major tasks planned for the next stage.

Descriptions should be sufficiently specific to demonstrate identifiable individual contribution.

- **References**

Provide a complete list of all sources cited in the report using a consistent recognised academic citation style.

APA and IEEE citation styles are preferred for this course.

Students should ensure that:

- every in-text citation has a corresponding reference;
- references listed are actually used in the report;
- external figures, tables, data, designs, technical specifications, software, APIs, libraries, or other resources are appropriately attributed where required; and
- reference information is sufficiently complete and accurate.

- **Appendices** (if applicable)

Include supplementary materials that support the main report but would interrupt the flow if included in the main body. Examples may include:

- additional test data;
- extended calculations;
- code excerpts;
- detailed technical specifications;
- additional diagrams;
- screenshots or logs;
- receipts or detailed cost records; and
- other supporting project evidence.

Appendices should be clearly labelled and referred to from the main body where relevant.

Relationship with the Marking Scheme

The **formal report sections above and the marking-scheme categories are not intended to correspond one-to-one**.

The marking scheme assesses the quality of relevant content wherever it is appropriately presented in the report. For example:

Marking-Scheme Category	Principal Evidence Normally Found In
Abstract / Executive Summary	Abstract
Review of Current System / Market Environment	Primarily Literature Review; relevant Introduction content may also contribute
Detailed Requirement Specifications	Primarily Literature Review and derived requirements; relevant Design content may also contribute
Detailed Design Specifications	Primarily Design
Current Progress	Primarily Implementation, preliminary testing, and relevant Discussion
Scheduling and Resource Allocation Review	Scheduling and Resource Allocation
Overall Impression	The report as a whole

The Interim Report is assessed according to the following weighting:

Assessment Area	Weighting within the Interim Report
Abstract / Executive Summary	5%
Review of Current System / Market Environment	10%
Detailed Requirement Specifications	25%
Detailed Design Specifications	25%
Current Progress	15%
Scheduling and Resource Allocation Review	10%
Overall Impression	10%
Total	**100%**

Where an assessed element is materially missing or contains no meaningful relevant content, the corresponding assessment category may be assessed within the **Inadequate** band.

Individual Self-Reflection Report (5%)

The Individual Self-Reflection Report is an **individual assessment** intended to evaluate each student's role, contribution, learning, and reflection on the experience of completing the project as a member of the group. The report should be written individually and should reflect the student's own project experience. It should not merely repeat the general description of the project.

Students are expected to provide **specific and credible examples** of their work and experiences rather than relying on broad statements such as "helped with programming", "worked on the hardware", or "learned teamwork".

Where appropriate, students should refer to concrete project evidence such as completed tasks, technical outputs, design work, source code, prototype development, test results, documentation, meeting records, Logbook entries, or other identifiable contributions.

The report should normally address the following areas.

- **Role and Responsibilities**

  Describe the student's role and responsibilities within the project group. The discussion should explain:

  - the main responsibilities assigned to the student;
  - the technical and/or non-technical areas for which the student was primarily responsible;
  - how these responsibilities related to the overall project; and
  - whether the student's role or responsibilities changed during the project and, if so, why.

  The description should distinguish between the student's expected responsibilities and the work that was actually completed.

- **Contributions to the Project**

  Describe and evaluate the student's actual contributions to the project.

  This should include specific tasks, technical work, decisions, outputs, or deliverables completed by the student. Students should explain what they personally contributed and how those contributions supported the development or completion of the overall project.

  Where a task was completed collaboratively, the student's own contribution should be identified as clearly as reasonably possible. Relevant evidence may be referenced where appropriate to support the reflection.

- **Skills Acquired from the Project**

  Reflect critically on the knowledge, skills, and experience gained through the project.

  Students should go beyond simply listing skills acquired. A strong reflection should explain:

  - what was learned;
  - how it was learned;
  - why it was important to the project;
  - how the student's understanding or ability changed; and
  - how the experience may influence future study or professional work.

Where difficulties, mistakes, unsuccessful approaches, or changes of direction contributed to learning, these may also form an important part of the reflection.

- **Team Coordination Efficiency**

  Evaluate the effectiveness of the project group from the student's perspective. The discussion may be considered:

  - division of responsibilities;
  - communication among group members;
  - coordination of tasks;
  - attendance and participation;
  - decision-making;
  - management of disagreements or technical differences;
  - sharing of information and project resources;
  - ability to meet milestones and deadlines; and
  - overall effectiveness of collaboration.

  Students should identify both strengths and areas for improvement where appropriate. The discussion should be constructive and supported by specific examples rather than consisting only of general statements such as "the teamwork was good" or "communication could be improved".

  Where significant concerns regarding individual contribution or possible free-riding exist, students should follow the Peer Evaluation and Free-Rider Reporting procedures specified in the relevant subsection of these Guidelines. The Self-Reflection Report alone does not constitute formal reporting of a free-rider case.

- **Personal Log or Supporting Evidence (if applicable)**

  Students may include or refer to a personal activity log or other supporting evidence where this helps demonstrate their contribution and development throughout the project.

Overall Quality of the Report

The Individual Self-Reflection Report should be:

- clearly organised;
- professionally presented;
- concise but sufficiently detailed;
- written in an appropriate academic and professional tone;
- internally consistent; and
- focused on genuine reflection rather than merely describing project activities.

The quality of reflection is important. Higher-quality work should demonstrate critical consideration of the student's own performance, learning, decisions, difficulties, and development, rather than simply providing a chronological list of tasks completed.

Assessment Weighting

The Individual Self-Reflection Report contributes **5% of the overall course grade***.

Peer Evaluation(s) (5%)

Peer Evaluation is an **individual assessment** intended to encourage appropriate individual contribution, responsibility, and cooperation throughout the two-semester group project.

A total of **five Peer Evaluations** will be conducted during the course. **Each Peer Evaluation contributes a maximum of 1% of the overall course grade**, giving a total weighting of 5%.

The five evaluations will normally be conducted as follows:

Semester	Evaluation Period	Maximum Course Mark
Semester 1	November 2026	1%
Semester 2	January 2027	1%
Semester 2	February 2027	1%
Semester 2	March 2027	1%
Semester 2	April 2027	1%
	Total	**5%**

Students are required to complete and submit **each Peer Evaluation individually** within the announced submission period.

- **Peer Evaluation Criteria**

  For each evaluation, students will be asked to assess their own contribution and the contributions of other members of the group in the following areas:

  - Performance on Tasks; and
  - Degree of Cooperation.

  The following rating scale will be used:

Performance Level	No Effort*	Inadequate*	Borderline	Satisfactory	Good	Excellent*
Rating	0	1	2	3	4	5

  Students must provide appropriate written comments when assigning a rating of 0, 1, or 5 in either assessment area.

  Comments should be specific and, where possible, supported by relevant examples or evidence. Students should avoid unsupported, personal, emotional, or retaliatory comments.

  Peer Evaluation should be completed independently and honestly. Students should not agree in advance to give one another particular ratings, exchange ratings for mutual benefit, or attempt to influence another member's evaluation.

- **Submission and the 1% Assessment Mark**

  Each Peer Evaluation is worth a **maximum of 1 percentage point of the overall course grade**.

  Submission of the student's own Peer Evaluation is required in order to be eligible for the mark associated with that evaluation.

If a student fails to submit a required Peer Evaluation by the applicable deadline, the student will receive 0% for that Peer Evaluation. The lost 1 percentage point will not be restored based on favourable ratings received from other group members.

Accordingly, a student who fails to submit one of the five Peer Evaluations can obtain a maximum of 4% out of the 5% Peer Evaluation component, regardless of how positively other group members evaluate that student's contribution.

Submission alone does not prevent the Peer Evaluation results from being considered when evaluating the student's actual contribution. Ratings and written comments received from group members may be reviewed together with other available evidence of individual participation and project work.

- **Use of Peer Evaluation Results**

Peer Evaluation serves both as an assessed component and as one source of evidence concerning individual contribution to the project.

The supervisor may consider Peer Evaluation results together with other relevant evidence, including but not limit to:

- observations during project sessions;
- individual progress demonstrated during supervisor consultation;
- project records and Logbook entries;
- identifiable technical or documentation outputs;
- Individual Self-Reflection Reports;
- presentation performance;
- source-code or version-control records;
- meeting records;
- photographs, screenshots, testing records, or other project artefacts; and
- other appropriate evidence.

A single low rating does not automatically establish inadequate contribution or a free-rider case. Similarly, consistently high peer ratings do not override clear evidence of insufficient individual contribution.

Where significant and credible differences in contribution are identified, Peer Evaluation results may contribute to adjustments in the relevant individual assessment components.

Free-Rider Reporting and Intervention

A **free rider** generally refers to a group member who makes substantially insufficient contribution to the group project while benefiting from work completed by the other members.

Peer Evaluation provides one mechanism for documenting concerns about unequal contribution. However, because a formal free-rider case may lead to substantial consequences for an individual student, **Peer Evaluation alone is not sufficient to initiate formal intervention**.

Formal Reporting Requirements

Where group members wish the project supervisor and/or Course Coordinator to formally investigate and intervene in a suspected free-rider case, **both of the following are required**:

- **The concern must be recorded through the relevant Peer Evaluation(s); and**
- **A formal email must be sent to the supervisor using the student's College email account.**

Both requirements should normally be satisfied. A concern mentioned only in a Peer Evaluation, Individual Self-Reflection Report, verbal discussion, instant message, or informal communication will not normally be treated as a formal request for free-rider intervention.

The formal email should identify the concern clearly and provide sufficient information for the supervisor to understand the situation. Where possible, relevant evidence should also be provided or identified. Examples of supporting evidence may include:

- agreed task allocations;
- meeting minutes or records;
- Logbook entries;
- communication records;
- source-code or version-control histories;
- shared-document revision histories;
- photographs or screenshots;
- records of missed agreed work or deadlines;
- evidence of repeated non-participation; and
- other relevant project records.

Timely Reporting

Concerns should be raised **as soon as a persistent contribution problem becomes apparent**.

Students should not wait until the final assessment period to report a problem that has existed for a substantial period of time, particularly where earlier reporting could reasonably have allowed the supervisor to review the situation and provide guidance or intervention.

Reporting a concern does not automatically establish that another student is a free rider. The supervisor and/or Course Coordinator will review the information available and may seek explanations or additional evidence from the students concerned before determining whether further action is appropriate.

Free-Rider Mark Adjustment

Where the supervisor and/or Course Coordinator determines, based on the available evidence, that a student has made substantially inadequate contribution to the group project, the student's marks in the relevant **individual and/or contribution-related assessment components** may be substantially reduced.

The extent of any adjustment will depend on the seriousness and duration of the contribution problem and the available evidence.

In serious cases, the assessment outcome may result in a Fail grade for CCIT4080.

As stated in Section (1), failing CCIT4080 generally means that the student has not fulfilled an important graduation requirement of the AENG programme, subject to the applicable programme and College regulations.

Responsibility of the Remaining Group Members

A contribution problem involving one or more members does not remove the group's responsibility to manage the project and submit the required assessments.

Where reasonably possible, the remaining members should continue to:

- maintain project progress;
- review and redistribute essential tasks where necessary;
- communicate significant risks to the supervisor;
- preserve appropriate evidence of task allocation and contribution; and
- ensure that required group submissions are completed by the applicable deadlines.

The existence of a confirmed or suspected free-rider case does not automatically justify incomplete or late group submissions.

Project Final Report (20%)

The Project Final Report is the complete technical documentation of the project and should provide a coherent, evidence-based account of the project from problem identification and investigation through design, implementation, testing, evaluation, and final conclusions.

Each group should produce a professionally formatted report of **at least 2000 words**.

The Final Report should reflect the **final state of the project**. It should therefore update and refine relevant material developed for the Project Proposal and Interim Report rather than simply combining or reproducing earlier submissions.

Materials from the Interim Report may be incorporated where relevant, but they should be reviewed and revised to reflect:

- feedback received during the course;
- changes in project scope or objectives;
- updated requirements;
- final design decisions;
- actual implementation;
- final testing and results;
- changes in schedule or resource usage; and
- the final conclusions that can reasonably be supported by the completed work.

The following sections are expected. Subsections may be added or adapted according to the nature of the project, provided that the required technical and assessment content is clearly presented.

- **Front Page**

  The Front Page should include the following information in a clear and professional hierarchy:
  - College Name;
  - Programme Name;
  - Course Code and Course Name;
  - Class Number and Group Number;
  - Project Title;
  - Assessment Item, i.e., **Project Final Report**;
  - group members' names and Student IDs (SIDs);
  - supervisor's name; and
  - submission date.

- **Abstract**

  Provide a concise summary of the **completed projec**t. The Abstract should normally summarise:
  - the problem being addressed;
  - the proposed and implemented solution;
  - the principal methodology or technical approach;
  - major implementation outcomes;
  - key testing and evaluation results;
  - significant limitations; and
  - the overall conclusion of the project.

Unlike the Interim Report, the Final Report Abstract should focus on the actual outcomes achieved, rather than planned or remaining work.

Claims regarding functionality, accuracy, performance, effectiveness, reliability, or achievement should accurately reflect the evidence presented in the report.

- **Acknowledgement (if applicable)**
  Acknowledge relevant assistance, facilities, resources, technical support, or other contributions received from persons or organisations outside the project group.

- **Table of Contents**
  Provide a properly structured Table of Contents showing the report headings and corresponding **starting page numbers**.

  The section titles and numbering shown in the Table of Contents should match the headings used in the report.

- **List of Figures**
  Provide a numbered list of all figures with their captions and corresponding page numbers.

  All figures should be appropriately numbered, captioned, referred to in the main text, and attributed where external material is used.

- **List of Tables**
  Provide a numbered list of all tables with their titles and corresponding page numbers.

  Tables should be numbered consistently, clearly titled, and appropriately referred to in the main text.

- **Introduction**
  Introduce the project and establish the context necessary for understanding the completed work.

  The Introduction should normally include:
  - project background and context;
  - the specific problem or engineering need addressed;
  - project goals and objectives;
  - final project scope;
  - relevant constraints or assumptions;
  - important theories, concepts, or terminology required to understand the project; and
  - an overview of the overall development process and report structure.

  The objectives presented should correspond to the final agreed project objectives.

  Where the scope, objectives, or technical direction changed during the project, the significant changes should be explained appropriately.

- **Literature Review**

  Review and critically analyse relevant literature, existing systems, products, services, technologies, theories, designs, methods, or other prior work that informs the project.

  The scope of the Literature Review will depend on the nature of the project. It may include:
  - academic literature;
  - existing commercial products or services;
  - comparable engineering systems or prototypes;
  - relevant theories and concepts;
  - engineering designs or reference architectures;
  - algorithms and technical methods;
  - hardware or software technologies;
  - standards and specifications;
  - datasheets or other authoritative technical documentation; and
  - other relevant sources.

  The Literature Review should not merely summarise existing work. It should demonstrate how the review informed the development of the project by:
  - comparing relevant alternatives;
  - identifying strengths and limitations;
  - identifying gaps or unresolved problems;
  - supporting technical assumptions;
  - justifying requirements; and
  - informing subsequent design and engineering decisions.

  Where appropriate, comparison tables or other structured forms should be used to compare existing systems, products, methods, or technical approaches.

- **Project Requirements**

  Relevant findings from the Literature Review and problem investigation should lead to a clear set of final project requirements.

  These may include:
  - Functional Requirements (FRs): what the proposed system or solution is required to do; and
  - Non-Functional Requirements (NFRs): relevant performance, reliability, usability, safety, cost, compatibility, resource, environmental, or other constraints.

  Requirements should be sufficiently clear and, where reasonably possible, measurable and testable.

  Where numerical targets or thresholds are specified, students should provide an appropriate rationale based on literature, technical specifications, calculations, preliminary investigation, user requirements, experiments, or other reasonable evidence.

  Where useful, requirement identifiers such as FR-01, FR-02, NFR-01, NFR-02, etc. should be used consistently throughout the report.

  The Final Report should demonstrate an appropriate relationship between:
  Problem and Objectives → Literature / Existing Solutions → Derived Requirements → Design → Implementation → Testing and Evaluation

- **Design, Methodology, and Implementation**

  This section should provide a technically detailed account of the solution developed, the methods used to develop and evaluate it, and the final implementation.

  Depending on the project, the section may be divided into the following subsections.

  - **Design**

    The Design should explain **what the final system or solution was designed to do and how its major components work together**.

    Depending on the nature of the project, appropriate content may include:

    - overall system architecture;
    - hardware and software subsystems;
    - circuit design;
    - mechanical or structural design;
    - software architecture;
    - algorithms and processing logic;
    - databases and data structures;
    - user-interface design;
    - communication interfaces;
    - control logic;
    - workflow or state transitions;
    - component selection;
    - integration between subsystems; and
    - etc.

    Appropriate engineering representations should be used where applicable, such as:

    - system block diagrams;
    - circuit schematics;
    - mechanical drawings;
    - wiring diagrams;
    - flowcharts;
    - state diagrams;
    - software architecture diagrams;
    - sequence diagrams;
    - interface diagrams; and
    - other relevant technical figures.

    Major design choices should be technically justified. Where reasonable alternatives were considered, students should explain the relevant comparison and why the final approach was selected.

    The final design should be clearly linked to the project requirements established earlier in the report.

- **Methodology**

  The Methodology should explain how the system or solution was developed, investigated, verified, and evaluated.

  Depending on the project, this may include:
  - development or prototyping approach;
  - experimental methodology;
  - technical decision-making methods;
  - data-collection procedures;
  - calibration procedures;
  - algorithm-development methods;
  - component or approach-selection methods;
  - integration strategy;
  - verification methods;
  - testing procedures;
  - evaluation metrics; and
  - acceptance or success criteria.

  Where alternative methods were considered, students should briefly compare them and justify the adopted approach.

  The Methodology should not merely repeat the Design or list the technologies used. It should explain the **process and reasoning used to develop and evaluate the project**.

- **Implementation**

  The Implementation should document **what was actually developed, constructed, programmed, configured, integrated, and verified**.

  Students should clearly distinguish among:
  - completed work; and
  - planned or remaining work.

  Depending on the project, appropriate evidence may include:
  - prototype photographs;
  - circuit and wiring records;
  - mechanical fabrication;
  - source code and software modules;
  - interface screenshots;
  - implemented algorithms;
  - hardware-software integration;
  - experimental setups;
  - configuration records;
  - measurement data;
  - test logs;
  - version-control records; and
  - other relevant project artefacts.

The Final Report should distinguish clearly between:

- features or functions successfully implemented;
- features only partially implemented;
- features removed or changed from the original plan; and
- features that could not be completed.

Where the final implementation differs from the original proposal or design, the changes and reasons should be explained.

Significant technical problems encountered during implementation should also be discussed, including investigation undertaken, solutions attempted, design modifications made, and the resulting outcomes.

- **Scheduling and Resources Allocation**

Review the management of the project across the full two-semester period.

The Final Report should **compare the original project plan with the actual project development**.

The schedule review should normally include:

- originally planned tasks and milestones;
- actual completion dates or status;
- significant delays or changes;
- causes of major differences;
- actions taken in response;
- changes in priorities or project scope; and
- overall effectiveness of the project schedule.

A planned-versus-actual schedule, revised Gantt chart, milestone table, or other suitable representation should be provided.

The report should also review the actual use of project resources, including where applicable:

- division of responsibilities among group members;
- laboratory and equipment usage;
- hardware and materials;
- software and computing resources;
- APIs, cloud services, or external services;
- expenditure and budget;
- procurement; and
- other significant resources.

Actual expenditure should be compared with the original or revised budget where applicable.

The purpose of this section is not merely to reproduce the original schedule and resource plan, but to **evaluate how effectively the project was actually managed**.

- **Testing, Evaluation, and Discussion**

  Testing and evaluation form a major part of the Final Report and should provide evidence for determining whether the project objectives and requirements have been achieved.

  - **Testing**

    This section should provide a detailed description of the testing process used to validate the project's solution.

    Students should develop and execute appropriate test cases, experiments, measurements, or other verification activities to evaluate the functionality and performance of the system or solution.

    For each significant test, the report should clearly identify, where applicable:
    - the functionality, component, subsystem, requirement, or objective being tested;
    - the test setup and conditions;
    - the procedure used;
    - the expected result or acceptance criterion;
    - the actual result obtained; and
    - whether the requirement or criterion was satisfied.

    Testing should, where appropriate, include both component- or unit-level testing and integrated system-level testing. The solution should also be evaluated under different relevant scenarios or operating conditions rather than only under a single ideal condition.

    Testing should be sufficiently systematic to support meaningful conclusions about the final project. Results should be presented using appropriate:
    - tables;
    - graphs;
    - measurements;
    - screenshots;
    - photographs;
    - logs;
    - statistical summaries; or
    - other relevant evidence.

    Successful tests alone should not be selectively reported where significant failures, limitations, or unexpected results were also observed.

  - **Evaluation**

    After testing, students should evaluate the results obtained and determine the extent to which the completed project has met its **original objectives, requirements, specifications, and success criteria**.

    The evaluation should compare actual results with the expected outcomes and identify, where applicable:

- requirements fully satisfied;
- requirements partially satisfied;
- requirements not satisfied;
- differences between expected and actual performance;
- limitations or failure cases identified during testing;
- limitations of the testing performed;
- reliability or consistency of the results; and
- factors affecting the validity or generalisability of the findings.

Where quantitative targets have been established, actual measured performance should be compared with those targets.

The evaluation should also consider the **overall effectiveness of the implemented solution and whether the final project actually addresses the original problem as intended**.

Students should distinguish between what is supported by evidence and what remains uncertain or has not been adequately tested. Where appropriate, a r**equirements-to-test/result traceability table** may be used to summarise whether individual requirements were achieved.

The evaluation should distinguish between:

- what the evidence demonstrates;
- what can reasonably be inferred from the evidence; and
- what remains uncertain or has not been tested.

- **Discussion**

The Discussion should provide a critical analysis of the project's outcomes based on the implementation, testing, and evaluation presented earlier in the report.

Students should discuss the strengths and weaknesses of the final solution and explain the significance of the results obtained.

The discussion may include:

- reasons for successful or unsuccessful outcomes;
- strengths and weaknesses of the final solution;
- important engineering trade-offs;
- technical limitations;
- implementation difficulties and their impact;
- differences between expected and actual performance;
- implications of the results;
- lessons learned from significant technical decisions;
- limitations of the project scope or methodology; and
- improvements that could be made if further development were undertaken.

The Discussion should demonstrate critical reflection rather than simply restating the test

results. Students should explain why particular outcomes occurred, what was learned from them, and what could reasonably be improved if the project were developed further.

Any proposed improvements should arise logically from the evidence, limitations, or lessons identified in the completed project.

- **Conclusion and Recommendations**

Provide an overall conclusion based on the completed project and the evidence presented in the report.

The Conclusion should:

- summarise the problem addressed;
- summarise the solution developed;
- state the extent to which the original objectives were achieved;
- identify the most important project outcomes;
- recognise significant limitations or unachieved objectives; and
- provide an evidence-based overall judgement of the project.

The Conclusion **should not introduce new technical results or evidence** that have not already been presented and discussed.

Where appropriate, recommendations may be provided regarding:

- design improvements;
- implementation changes;
- improved testing methods;
- deployment considerations;
- optimisation;
- alternative technical approaches; or
- other lessons arising from the project.

- **Future Scope**

Discuss reasonable opportunities for further development, improvement, or investigation beyond the completed project.

Future work should arise logically from the limitations, unresolved issues, results, or opportunities identified in the completed project, rather than simply listing additional desirable features. Where appropriate, students may discuss:

- unresolved technical limitations;
- performance improvements;
- additional functionality;
- wider testing;
- improved hardware or software;
- alternative approaches;
- larger-scale deployment;
- further research; or
- application of the work in other contexts.

Students should distinguish clearly between what was achieved in the current project and what is proposed only as future work.

- **Individual Progress**

Provide separate subsections for each group member summarising their contribution to the completed project. Each member's subsection should include:

- portrait photograph;
- full name and Student ID (SID);
- role or major responsibilities;
- significant tasks completed;
- concrete technical or project outputs produced; and
- contribution to the overall project outcome.

Descriptions should be sufficiently specific to demonstrate identifiable individual contribution and should be consistent with other project evidence such as the Logbook, Peer Evaluation(s), Individual Self-Reflection Report, and etc.

As this is the Final Report, the Individual Progress section should focus primarily on actual completed contribution, rather than planned future tasks.

- **References**

Provide a complete list of all sources cited in the report using a consistent recognised academic citation style.

APA and IEEE citation styles are preferred for this course.

Students should ensure that:

- every in-text citation has a corresponding reference;
- references listed are actually used in the report;
- external figures, tables, data, designs, technical specifications, software, APIs, libraries, or other resources are appropriately attributed where required; and
- reference information is sufficiently complete and accurate.

- **Appendices**

Include supplementary materials that support the main report but would interrupt the flow if included in the main body. Examples may include:

- additional test data;
- extended calculations;
- code excerpts;
- detailed technical specifications;
- additional diagrams;
- screenshots or logs;
- receipts or detailed cost records; and
- other supporting project evidence.

Appendices should be clearly labelled and referred to from the main body where relevant.

Relationship with the Marking Scheme

The **formal report sections above and the marking-scheme categories are not intended to correspond one-to-one**.

The marking scheme assesses the quality of relevant content and evidence wherever it is appropriately presented in the report and, where applicable, in the completed project itself. The relationship between the marking-scheme categories and the principal sources of evidence is generally as follows:

Marking-Scheme Category	Principal Evidence Normally Found In
Abstract	Abstract
Introduction	Introduction
Detailed Requirement Specifications	Primarily Literature Review / Project Requirements
Detailed Design Specifications	Primarily Design
Evaluation	Testing, Evaluation and Discussion
Conclusion	Conclusion and Recommendations
Appendices and References	References and Appendices, together with the appropriate use, citation, and supporting documentation of sources and supplementary materials throughout the report
Overall Impression	The report as a whole, including organisation, technical communication, logical structure, formatting, figures and tables, consistency, referencing, and professional presentation
Product	The actual completed prototype, system, solution, or other approved project output, supported by evidence presented particularly in Implementation and Testing, Evaluation and Discussion

In particular, **Product is an assessment category rather than a required report section**. Students should not create a separate chapter entitled "Product" merely to match the marking scheme.

The Final Report contributes **20% of the overall course grade**.

The corresponding assessment areas are:

Assessment Area	Weighting within the Final Report
Abstract	5%
Introduction	5%
Detailed Requirement Specifications	10%
Detailed Design Specifications	10%
Evaluation	20%
Conclusion	10%
Appendices and References	10%
Overall Impression	10%
Product	20%
Total	**100%**

Where an assessed element is materially missing or contains no meaningful relevant content, the corresponding assessment category may be assessed within the **Inadequate** band.

Project Deliverable (35%)

The Project Deliverable represents the **project's actual technical outcome and the supporting materials required to demonstrate, explain, and document it**.

The deliverable should demonstrate not only that a prototype, system, or solution has been produced, but also the quality of its design and implementation, its functionality relative to the project objectives and requirements, the development progress achieved throughout the project, and the contributions made by individual group members.

Depending on the nature of the project, the deliverable may include a hardware prototype, software application, integrated hardware-software system, experimental setup, or other approved technical output.

In addition to the actual prototype or system, each group is required to provide:

- **Project Demonstration Video(s)**;
- **A2-size Project Presentation Poster**; and
- other supporting materials or technical documentation where appropriate.

- **Project Demonstration Video**

  The project video should be **at least 5 minutes** long and clearly demonstrate the completed prototype or system.

  The video should normally include:

  - a brief introduction to the project and the problem being addressed;
  - the purpose and major features of the prototype or system;
  - an explanation of the main components or functions;
  - step-by-step guidance showing how the prototype, device, or application is used;
  - demonstration of the major functions of the system;
  - demonstration under representative operating scenarios where appropriate; and
  - relevant outcomes, limitations, or testing evidence where these assist in demonstrating the performance of the system.

  The demonstration should show the actual implemented system or prototype wherever reasonably possible. Planned functions, simulated outputs, mock-ups, or incomplete features should not be presented as completed functionality.

  The purpose of the video is primarily to **demonstrate and document the project deliverable**. It is therefore different from the longer Project Oral Presentation, which focuses on communicating the complete project process, technical decisions, testing, evaluation, and conclusions.A brief introduction to the prototype, explaining its purpose and key features.

- **A2-size Project Presentation Poster**

  Each group should prepare an A2-size presentation poster that provides a concise and visually effective summary of the project.

  Depending on the nature of the project, the poster may include:

  - project problem and objectives;
  - proposed solution;
  - system or solution architecture;
  - important design features;
  - methodology or development approach;
  - prototype or implementation photographs;
  - significant testing or evaluation results;
  - major project outcomes; and
  - key conclusions or limitations.

  The poster should allow a reader to understand the main idea and outcome of the project quickly without having to read the complete Final Report.

  The poster should therefore emphasise clear visual communication, making appropriate use of diagrams, figures, photographs, tables, and concise text rather than presenting large amounts of continuous prose.

- **Supporting Materials and Documentation**

  Where appropriate, groups should also provide supporting materials necessary to allow the project to be understood, demonstrated, reproduced, operated, or evaluated.

  Depending on the project, these may include:

  - user or operating instructions;
  - installation or setup instructions;
  - circuit diagrams or wiring information;
  - mechanical drawings;
  - source code or executable software;
  - README or configuration files;
  - bills of materials;
  - component specifications;
  - test records;
  - datasets or configuration information; and
  - other relevant technical documentation.

  The required supporting materials will depend on the nature and scope of the project.

- **Functionality and Quality of the Final Product**

  The final prototype or solution should be evaluated against the **project objectives, requirements, specifications, and intended use** established during the project.

  A technically ambitious project is not automatically considered successful if its major functions have not been implemented or demonstrated. Similarly, a comparatively focused project may perform strongly where the design is appropriate, implementation is technically sound, functionality is well demonstrated, and the resulting solution is thoroughly tested and evaluated.

  Where some intended functions could not be completed, the group should identify these limitations accurately rather than representing incomplete work as fully functional.

- **Project Development and Progress**

  The Project Deliverable assessment considers not only the final state of the prototype, but also the development work and progress demonstrated throughout the project.

  Relevant evidence may include:
  - progress demonstrated during project sessions;
  - prototype-development records;
  - design revisions;
  - implementation evidence;
  - testing records;
  - technical problem-solving;
  - project documentation;
  - Logbook records; and
  - other project artefacts.

  Substantial project work completed only immediately before the final submission, with little evidence of continual development, will negatively affect the relevant gradings.

- **Individual Contribution**

  A substantial proportion of the Project Deliverable is assessed individually. Each student is expected to demonstrate a fair, meaningful, and identifiable contribution to the development of the project.

  Individual assessment may consider:
  - technical contribution to design and implementation;
  - contribution to relevant project documentation;
  - contribution to testing, troubleshooting, and project development;
  - engagement and communication within the group;
  - progress demonstrated during project sessions;
  - Peer Evaluation(s);
  - Individual Self-Reflection Report;
  - Logbook and project records;
  - source-code, document, or version histories where relevant; and
  - other appropriate evidence of individual contribution.

Where work has been completed collaboratively, the assessment will consider the available evidence concerning each member's contribution.

Students should not assume that all group members will automatically receive identical individual marks. Where the available evidence indicates comparable levels of contribution, similar individual marks may be awarded.

- **Assessment Weighting**

The Project Deliverable contributes **35% of the overall course grade**.

The assessment is divided into:

Assessment Area	Weighting within the Course
Group-Based Assessment – First Examiner	10%
Group-Based Assessment – Second Examiner	5%
Individual Assessment	20%
Total	**35%**

The **Group-Based Assessment** considers the overall quality of the project as a collaborative engineering outcome. The principal areas include:

- Design and Idea;
- Implementation, Project Progress, and the Final Product; and
- Project Video and Documentation.

The Individual Assessment considers each student's contribution to the project. The principal areas include:

- Technical Contribution to Implementation;
- Contribution to Relevant Documentation; and
- Engagement and Communication in the Group.

Note: In general, if no significant differences are observed in in-class performance, Peer Evaluation(s), or Individual Self-Reflection Reports, students may expect similar individual marks to be awarded within the group.

Project Oral Presentation (15%)

Each group is required to prepare a **self-recorded Project Oral Presentation of approximately 30–40 minutes**.

The presentation should communicate the completed project to an **engineering audience who may have no prior knowledge of the project's development**, including the second examiner.

The presentation should provide a coherent account of the project rather than merely describing the final prototype. It should demonstrate the group's understanding of the problem, engineering decisions, development process, implementation, testing, evaluation, and final outcomes.

- **Presentation Content**

  The presentation should normally cover the major elements of the completed project, including:
  - the problem or engineering need being addressed;
  - project goals and objectives;
  - important findings from the background investigation or Literature Review;
  - major project requirements;
  - proposed solution and system design;
  - important technical decisions and their justification;
  - methodology or development approach;
  - implementation and project development;
  - major difficulties encountered and how they were addressed;
  - testing methodology and test cases;
  - testing and evaluation results;
  - strengths and limitations of the final solution;
  - conclusions and major lessons learned; and
  - appropriate future improvements or extensions.

  The presentation should include **a brief demonstration of the prototype or system**, focusing on the major functions and project outcomes.

  Materials from the Project Deliverable, including demonstration footage, diagrams, photographs, or testing evidence, may be reused where appropriate.

- **Presentation Style and Materials**

  The presentation should be **clear, concise, logically organised, and professionally presented**.

  Although some content will naturally overlap with the Project Final Report, students should not simply read the Final Report or reproduce large sections of report text on presentation slides.

  Presentation materials should support the oral explanation through appropriate use of:
  - system and technical diagrams;
  - figures and photographs;
  - tables and graphs;
  - test results;
  - screenshots;

- demonstration footage; and
- concise explanatory text.

The presentation should focus on the most important aspects of the project and communicate them at an appropriate technical level for an engineering audience.

- **Participation of Group Members**

All group members are expected to make a **meaningful contribution to the recorded presentation**.

The division of presentation content does not necessarily need to be equal in duration, but the allocation should reasonably reflect individual responsibilities, technical involvement, and the overall presentation structure.

Each student should be clearly identifiable during their part of the presentation.

Individual presentation performance may consider:

- clarity and effectiveness of explanation;
- organisation and delivery;
- appropriate use of technical terminology;
- demonstrated understanding of the material being presented;
- ability to explain the student's own project work and contribution; and
- overall professionalism of presentation.

Students should demonstrate understanding rather than simply reading prepared text.

- **Group and Individual Assessment**

The Project Oral Presentation contains both group-based and individual assessment.

The group-based assessment considers the overall quality and effectiveness of the presentation, including:

- selection and organisation of appropriate project content;
- communication of key material from the Final Report;
- demonstration of the prototype or system;
- quality and professionalism of presentation materials;
- coherence and consistency across different presenters; and
- overall effectiveness in communicating the completed project.

The individual assessment considers each student's own presentation performance and contribution to communicating the project.

- **Assessment Weighting**

  The Project Oral Presentation contributes **15% of the overall course grade**.

  The assessment is divided into:

Assessment Area	Weighting within the Course
Group-Based Assessment – First Examiner	5%
Group-Based Assessment – Second Examiner	5%
Individual Assessment	5%
Total	**15%**

Note: As with the project deliverable, if no significant differences are identified through performance observations, Peer Evaluation(s), or Individual Self-Reflection Reports, students may expect similar individual marks across the group.

Note: The presentation should be concise, focusing on key points. While it may include elements similar to the Final Report, it should be presented in a lively tone rather than simply reading out the Final Report. Materials from the **Project Deliverable** may be reused in the recording where appropriate.

Logbook (5%: 2.5% + 2.5%)

Each project group is required to maintain a **Logbook throughout the entire two-semester project period**.

The Logbook serves as a chronological record of the development of the project and should provide evidence of the group's activities, progress, discussions, decisions, problems encountered, and actions taken.

The Logbook will be submitted at the end of each semester:

- Semester 1 Logbook – 2.5%
- Semester 2 Logbook – 2.5%

The Logbook should be **maintained continuously throughout the project**, rather than reconstructed shortly before the submission deadline.

- **Activities and Milestones**

  The Logbook should provide a chronological record of significant project activities and milestones.

  Entries may include:

  - research and investigation;
  - design work;
  - component selection;
  - programming or development;
  - prototype construction;
  - experimentation;
  - integration;
  - testing;
  - troubleshooting;
  - documentation;
  - meetings;
  - supervisor consultation;
  - significant revisions; and
  - other relevant project activities.

  The Logbook should make it possible to understand how the project developed over time.

- **Date**

  Each activity, meeting, milestone, or significant project event should be associated with an appropriate date or period.

  Entries should be sufficiently regular to provide a credible record of project development throughout the semester.

- **Description of Work and Progress**

  Logbook entries should describe the work undertaken in sufficient detail.

  Depending on the activity, the description may include:
  - tasks completed;
  - work in progress;
  - findings or observations;
  - technical difficulties encountered;
  - troubleshooting performed;
  - results obtained;
  - changes made;
  - unresolved problems; and
  - planned follow-up actions.

  Entries should describe actual project activity and outcomes, rather than consisting only of brief statements such as "worked on coding" or "continued hardware development".

  Where appropriate, entries may refer to supporting evidence such as photographs, screenshots, diagrams, test results, source-code records, or other project artefacts.

- **Meeting Agenda and Minutes**

  Relevant group meetings should be documented through appropriate meeting records.

  Meeting records should normally include:
  - meeting date;
  - members present, where appropriate;
  - major agenda items;
  - important discussions;
  - decisions made;
  - tasks or action items assigned;
  - responsible member(s); and
  - expected follow-up or completion dates where relevant.

  Records of supervisor consultation may also be included where appropriate.

- **Key Decisions**

  Important decisions affecting the direction or development of the project should be recorded.

  Examples may include:
  - changes in project scope;
  - selection between alternative technical approaches;
  - component or technology choices;
  - design modifications;
  - changes to implementation;
  - changes to testing methods;
  - schedule revisions;

- resource or budget decisions; and
- responses to significant technical problems.

Where appropriate, the Logbook should briefly record the reason or evidence supporting the decision, rather than documenting only the final choice.

- **Individual Responsibilities and Contribution**

Although the Logbook is maintained as a group record, it should make individual responsibilities and contributions reasonably identifiable where appropriate.

Tasks, action items, development activities, or project responsibilities should therefore identify the relevant member(s) where this improves the accuracy and usefulness of the project record.

The Logbook may be considered together with other evidence when reviewing individual contribution to the project.

- **Quality and Authenticity of the Logbook**

The purpose of the Logbook is to provide a credible record of the actual project-development process.

A strong Logbook should demonstrate:

- regular and continuous project activity;
- meaningful descriptions of progress;
- clear records of important discussions and decisions;
- evidence of problems, troubleshooting, and subsequent action;
- appropriate tracking of tasks and milestones; and
- consistency with the project progress demonstrated elsewhere.

A Logbook consisting mainly of brief, repetitive, or retrospective entries provides limited evidence of genuine project management and development.

- **Assessment Weighting**

The Logbook contributes 5% of the overall course grade.

Assessment Area	Weighting within the Course
Semester 1 Logbook	2.5%
Semester 2 Logbook	2.5%
Total	**5%**

Each semester's Logbook is assessed as a whole based principally on the quality and completeness of its content, including:

- Activities and Milestones;
- Date;
- Description of tasks, findings, difficulties, and progress;
- Meeting Agenda and Minutes; and
- Key Decisions.

(7) Academic Honesty

Students are advised to read the University's guidelines on academic honesty (https://tl.hku.hk/plagiarism/). The definition of plagiarism includes (but is not limited to) copying the whole or parts of any course assessment. Note that the College places very high importance on academic integrity, and adopts a policy of zero tolerance on cheating and plagiarism. Any related offence shall lead to disciplinary actions, including termination of studies.

The policy of the College and this course about the use of artificial intelligence tools is very clear:

- Plagiarism is not allowed at any time for any type of assessment. The passing off of work produced by AI as one's own without acknowledgment is regarded as plagiarism and will be penalized. Students must disclose the use of AI tools in their assessments and provide adequate documentation of the usage process. Instructors reserve the right to question the assessments submitted and may require students to explain their methodology, including showing detailed steps in obtaining results from Gen-AI tools.